Quantitative Portfolio Math & Reasoning

1. 200-Day Moving Average via QR Decomposition

Typically, a 200-Day Moving Average (200DMA) is calculated as a simple mean: SMA = sum(P_i)/200. However, strictly speaking, finding the mean is equivalent to finding the scalar 'c' that minimizes the least squares error || y - A*c ||^2, where 'y' is the 200x1 vector of daily closing prices and 'A' is a 200x1 column vector of ones.

Step-by-step QR Decomposition:

1. We define A as a 200x1 matrix of 1s.
2. We factor A into an orthogonal matrix Q (200x1) and an upper triangular matrix R (1x1).
   Since A is just ones, Q = (1/sqrt(200)) * A, and R = sqrt(200).
3. The least squares solution for the moving average 'c' is given by c = R^(-1) * Q^T * y.
4. Q^T * y is simply (1/sqrt(200)) * sum(y). Multiplying by R^(-1) = 1/sqrt(200) yields sum(y)/200.

Thus, the QR decomposition perfectly derives the exact 200DMA!

2. Eigenvalues and Eigenvectors (PCA of Portfolio)

To understand the 'hidden forces' driving the portfolio, we calculate the Eigenvalues and Eigenvectors of the covariance matrix of daily asset returns.

Step-by-step Process:

1. Let 'X' be the T x N matrix of daily returns for N assets over T days.
2. We compute the N x N covariance matrix: C = (X^T * X) / (T - 1).
3. We solve the characteristic equation det(C - lambda*I) = 0 to find the Eigenvalues (lambda).
4. The sum of all Eigenvalues represents the total variance of the portfolio.
5. For each Eigenvalue, we find its corresponding Eigenvector 'v', satisfying C*v = lambda*v. These vectors tell us the 'direction' of the hidden market factors (e.g., General Market Trend, Sector Trend).

Your Portfolio's Top 5 Principal Components (Hidden Factors):

Factor 1 Eigenvalue explains 42.58% of your portfolio's variance.
Factor 2 Eigenvalue explains 18.29% of your portfolio's variance.
Factor 3 Eigenvalue explains 12.72% of your portfolio's variance.
Factor 4 Eigenvalue explains 5.29% of your portfolio's variance.
Factor 5 Eigenvalue explains 4.64% of your portfolio's variance.

What do these statistical factors mean in reality?

Because these factors are generated purely by the Eigenvectors of your covariance matrix, they don't have hardcoded names. However, quantitative analysts typically interpret them as follows:

Factor 1: The General Market Direction (Macro Trend)
This usually explains 60% to 80% of variance. It captures the reality that most assets move with the broader market. If this factor explains >80% of your variance, your portfolio is acting as a single highly correlated unit.

Factor 2: Asset Class & Sector Divides (e.g., Stocks vs. Gold)
Usually captures how different types of assets react to stress (e.g., when equities fall, gold rises).

Factor 3: Industry-Specific Shocks
Groups companies in the same industry. For example, infrastructure policy changes affecting PSU stocks while leaving tech stocks untouched.

Factor 4: Value vs. Growth / Interest Rate Sensitivity
Captures how stocks react to borrowing costs. High-growth tech companies react differently to interest rate hikes than cash-rich dividend-paying companies.

Factor 5: Company-Specific Noise (Idiosyncratic Risk)
Picks up specific, isolated events like a single company reporting bad earnings, which has almost zero correlation with the rest of your portfolio.

3. Asset-Specific Quantitative Metrics

REC Limited:

* QR-Derived 200-Day Average: Rs. 356.42
* Current Price: Rs. 354.30 (Distance: -0.60%)
* Volatility (Standard Deviation): 40.09%
* Sharpe Ratio: -0.42
* Composite Risk Score: 72.41/100

Jio Financial Services:

* QR-Derived 200-Day Average: Rs. 288.44
* Current Price: Rs. 246.37 (Distance: -14.59%)
* Volatility (Standard Deviation): 30.62%
* Sharpe Ratio: -0.80
* Composite Risk Score: 70.06/100

IRCTC:

* QR-Derived 200-Day Average: Rs. 656.58
* Current Price: Rs. 539.55 (Distance: -17.82%)
* Volatility (Standard Deviation): 26.36%
* Sharpe Ratio: -1.38

Quantitative Portfolio Math & Reasoning

* Composite Risk Score: 69.26/100

Eternal (Zomato):

* QR-Derived 200-Day Average: Rs. 289.85
* Current Price: Rs. 247.03 (Distance: -14.77%)
* Volatility (Standard Deviation): 38.87%
* Sharpe Ratio: 0.32
* Composite Risk Score: 66.04/100

Coal India:

* QR-Derived 200-Day Average: Rs. 397.22
* Current Price: Rs. 481.45 (Distance: 21.21%)
* Volatility (Standard Deviation): 26.68%
* Sharpe Ratio: 0.23
* Composite Risk Score: 57.06/100

Tata Steel:

* QR-Derived 200-Day Average: Rs. 180.05
* Current Price: Rs. 211.36 (Distance: 17.39%)
* Volatility (Standard Deviation): 29.27%
* Sharpe Ratio: 0.39
* Composite Risk Score: 55.68/100

BEL:

* QR-Derived 200-Day Average: Rs. 411.60
* Current Price: Rs. 431.30 (Distance: 4.79%)
* Volatility (Standard Deviation): 34.46%
* Sharpe Ratio: 0.89
* Composite Risk Score: 54.72/100

Silver BeES:

* QR-Derived 200-Day Average: Rs. 181.68
* Current Price: Rs. 227.50 (Distance: 25.22%)
* Volatility (Standard Deviation): 44.60%
* Sharpe Ratio: 1.29
* Composite Risk Score: 54.15/100

Motilal Nasdaq 100 ETF:

* QR-Derived 200-Day Average: Rs. 229.02
* Current Price: Rs. 299.17 (Distance: 30.63%)
* Volatility (Standard Deviation): 25.81%
* Sharpe Ratio: 1.31
* Composite Risk Score: 52.25/100

Quantitative Portfolio Math & Reasoning

Nexus Select Trust:

* QR-Derived 200-Day Average: Rs. 154.27
* Current Price: Rs. 155.29 (Distance: 0.66%)
* Volatility (Standard Deviation): 17.74%
* Sharpe Ratio: 0.48
* Composite Risk Score: 38.91/100